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Low-Frequency DAS for Cement Quality Evaluation in Horizontal Wells

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Abstract

In hydraulic fracturing, the cement behind the casing is intended to isolate each fracturing stage during the plug-and-perf fracturing process. However, compromised cement bonding can lead to unintended behind-casing fluid pathways, significantly reducing isolation performance. This study introduces a novel method leveraging low-frequency distributed acoustic sensing (LF-DAS) measurements, verified by external pressure gauge data, to quantitatively determine annular pressure communication and cement quality along horizontal monitor wells. This method is verified with field measurements obtained from a Bakken multi-well pad instrumented with permanent fiber-optic cables and external pressure gauges. LF-DAS strain signals consistently showed heel-ward-migrating patterns synchronized with corresponding pressure increases, indicating fluid migration along partially debonded cement-formation interfaces. A numerical model simulating one-dimensional pressure diffusion with spatially variable hydraulic diffusivity successfully matched observed strain and pressure signals. The strain signals were attributed to radial-to-axial deformation via Poisson's effect, induced by annular pressure buildup. This integrated analysis demonstrates the feasibility and practical value of using cross-well LF-DAS for pre-stimulation cement quality evaluation. Crucially, while external gauges were used in this study to ground-truth the physical mechanism, we establish the workflow that future deployments can rely solely on fiber-optic sensing to detect and quantify leakage. The developed method offers a highly cost-effective alternative to conventional cement bond logging and enables proactive adjustments to completion designs before the monitor wells are stimulated.

Introduction

The widespread development of unconventional reservoirs has transformed the oil and gas industry, driven by advances in horizontal drilling and hydraulic fracturing technologies. In these operations, multistage hydraulic fracturing along extended laterals has become a standard practice, often completed using the plug-and-perf method. This completion design relies on a combination of cement, mechanical plugs, and perforation clusters to isolate and stimulate discrete sections of the wellbore in stages. Stage isolation, which prohibits stimulation fluid from restimulating hydraulic fractures created by the previous stages, is critical for ensuring stimulation efficiency, directly impacting production performance (e.g., Cramer and Zhang, 2021; Schumann and Jin, 2022; Haustveit et al., 2024).

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Successful stage isolation depends heavily on the integrity of the cement sheath between casing and formation. Inadequate zonal isolation due to poor cement bonding can result in inter-stage leakage, pressure communication, or ineffective stimulation. The most common method for investigating cement quality is through cement bond logging (CBL), which uses ultrasonic tools to assess the acoustic impedance of the annular materials and evaluate cement coverage (Timonin et al., 2014; Wang et al., 2016). However, running CBL tools is expensive and rarely applied in unconventional wells.

Recent advancements in distributed fiber-optic sensing (DFOS) present a promising approach for monitoring wellbore integrity and assessing stage isolation conditions (e.g., Ugueto et al., 2015; White et al., 2021). Nevertheless, most diagnoses focus on detecting stage leakage using DFOS measurements taken at stimulated wells (in-well), which necessitates the permanent installation of sensing cables behind the casing (e.g., Raterman et al., 2019). Mjehovich et al. (2024) employed cross-well strain signals (Jin and Roy, 2017) through low-frequency distributed acoustic sensing (LF-DAS) to quantitatively assess stage isolation of an offset well, by estimating the far-field reactivation of hydraulic fractures from earlier stages during the ongoing stage stimulation. However, this technique cannot differentiate whether the leakage results from cement or plug failure.

Additionally, many of the DFOS methods mentioned earlier primarily concentrate on diagnosing the wells during stimulation. Ideally, recognizing poor cement quality ahead of stimulation allows for adjustments to completion designs, potentially minimizing the effects of leakage behind the casing between stages. A recent study by Haustveit et al. (2024) analyzed cross-well LF-DAS signals prior to the stimulation of the monitor well, and compared these with results from cement bond logs. They identified a specific LF-DAS signal indicative of poor cement quality in the monitor well. This finding presents an opportunity to assess the cement quality of an unconventional well before its stimulation. Nonetheless, their interpretation is largely qualitative and does not explore the quantitative aspects of the signal.

This paper presents a novel case study of analyzing co-located external pressure gauges and LF-DAS data deployed in monitor wells adjacent to a treatment well. Notably, the monitor wells themselves were not stimulated. By combining pressure measurements with distributed strain measurements, we demonstrate how LF-DAS signals can be used to quantitatively evaluate the pressure communication—specifically locating areas of poor cement quality along the monitor wells prior to their own completion operations. The remainder of the paper is structured as follows. First, we describe the field data acquisition and highlight key strain signals observed in the LF-DAS measurements. Next, we present a conceptual model that characterizes fluid pressure migration through compromised cement zones, followed by a numerical model used to match the observed pressure and strain responses. We then discuss the implications of these findings for cement quality evaluation, and conclude with a summary of the modeling results and their relevance to future completion design strategies.

Field Data Description and Observations

The field data used in this study were acquired in the Bakken field, located within the Williston Basin across North Dakota and Montana. This region is a major unconventional oil-producing region primarily developed by XTO Energy (Liang et al., 2022). Key productive intervals in this asset include the Middle Bakken and Upper Three Forks formations, with additional productivity identified in the Lower Three Forks benches. The study pad includes one horizontal producer targeting Middle Bakken and two dedicated monitor wells: one in-plane Middle Bakken well (H well) and one slant Three Forks well (S well). The well trajectories are shown in Figure 1.

All three wells are equipped with fiber-optic sensing cables, permanently installed behind the casing and cemented in place. Each monitoring well is equipped with 15 external pressure gauges to monitor reservoir pressures, in addition to 5 external pressure gauges installed along the producer. Pressure measurements were continuously collected during the stimulation and production periods. It is critical to emphasize that during the stimulation of the producer, there was no fluid flowing inside the casing of the monitor wells (the S and H wells). Therefore, any migrating pressure or strain signals detected along the monitor wells are exclusively indicative of fluid movement within the external annular space between the casing and the formation. LF-DAS measurements were taken at the monitoring wells during the stimulation with a channel spacing of 1 m (note that, due to the gauge length, channel spacing does not equal spatial resolution for DAS data) and a temporal sampling interval of 1 s. This high-resolution configuration ensures that the rapid heel-ward migration of strain features is captured with sufficient detail for quantitative analysis. In addition to

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LF-DAS measurements, Rayleigh Frequency Shift Distributed Strain Sensing (RFS-DSS) measurements were acquired during a well interference test after 18 months of production. Readers can refer to Liang et al. (2022) for more detailed information about the dataset.

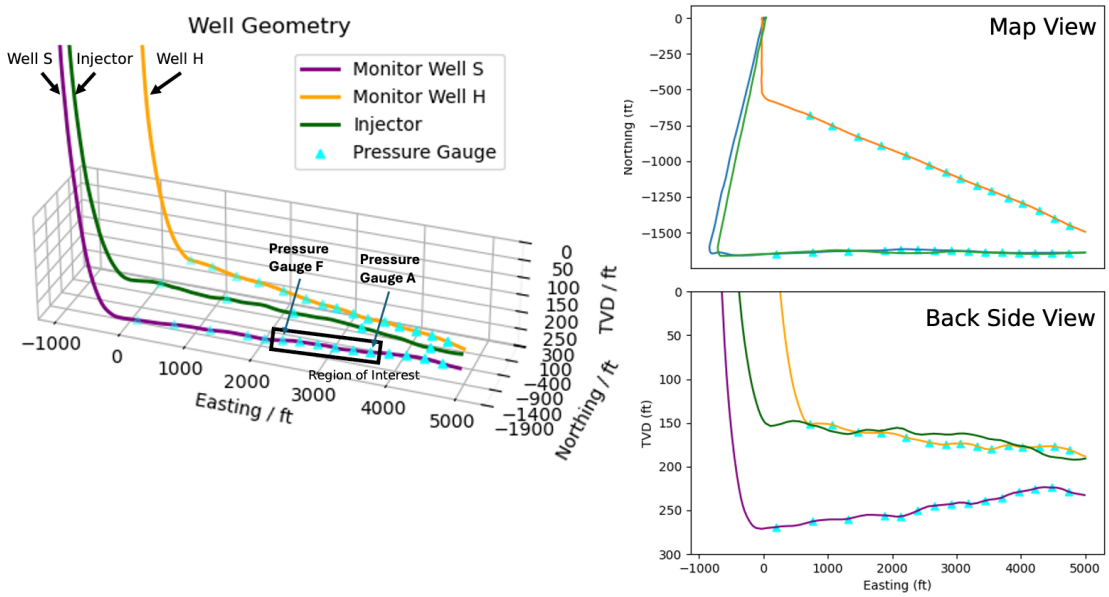


Fig. 1—Wellbore trajectories and pressure gauge locations. The 3D (left) and 2D (right) views show the spatial arrangement of the S well (purple), H well (yellow), and producer well (green), along with the external pressure gauges (cyan markers). The monitor wells target the Three Forks and Middle Bakken formations, respectively.

Figure 2 shows a typical cross-well strain rate signal observed at both monitoring wells during the stimulation of the producer. In addition to regular fracture hit characteristics (Ugueto et al., 2023), the LF-DAS signal also displays a near-linear feature migrating rapidly toward the heel. This signal is most prominent at the beginning and the end of each pumping stage. Meanwhile, pressure gauges located heel-ward of the fracture hit signals also observed a significant pressure perturbation, with the timing of the pressure variation aligning with the collocated LF-DAS measurements. Additionally, this comparison shows that the extensional strain rate corresponds with increasing pressure, whereas the compressional signals relate to decreasing pressure.

Moreover, we find that pressure communication in the pressure gauge data is much stronger in the heel-ward direction, with a notable decrease in magnitude toward the toe. The LF-DAS measurements reveal that the linear feature propagates more prominently towards the heel, whereas the signal's propagation towards the toe is disrupted and ends at the locations impacted by prior fracture hits.

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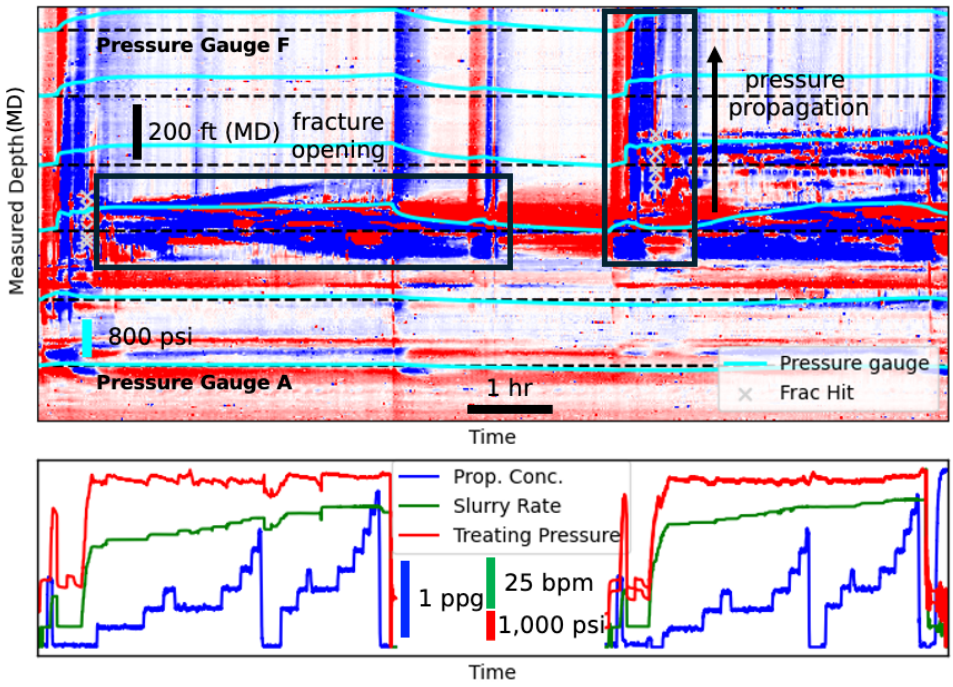


Fig. 2—LF-DAS and pressure gauge data showing heel-ward migrating strain change signal and pressure propagation during hydraulic stimulation. For the upper panel, the background is the waterfall plot of LF-DAS measurement from the S well, with cyan and black lines indicating the co-located pressure gauge readings. Black dashed lines indicate the locations of those external pressure gauges. The black box in the upper panel shows the pressure propagation signal captured by LF-DAS and pressure gauges. The lower panel shows the pumping curve during the two stages of interest: the blue curve shows the proppant concentration, the green curve shows slurry rate, and the red curve shows the treating pressure measured at the wellhead.

We interpret the linear migrating feature observed in the LF-DAS measurements in Figure 2 as being linked to pressure migration along the monitoring wellbore. This pressure migration is likely a result of poor cement quality in the monitoring well. As pressure builds in the annular space, it compresses the casing and the sensing cable connected to it radially, which induces axial extension strain due to Poisson's effect, causing the strain response captured by LF-DAS.

To quantitatively compare the strain rate with the pressure measurement, we convert the DAS signals from optical phase change to mechanical strain rate using the equation (Lindsey et al., 2020):

$$\varepsilon = \frac{\lambda}{4\pi n L_G \zeta} \Delta\Phi \tag{1}$$

where the incident wavelength λ = 1550 nm, refractive index n = 1.4682, and gauge length L_G = 4.08 m; the scalar coefficient ζ = 0.79 accounts for stress-induced birefringence, which governs the optical phase shift in response to axial strain as measured by the LF-DAS.

Subsequently, we calculated the temporal gradient of the pressure measurements to align better with the nature of the strain rate measurements. Figure 3 illustrates the comparison between the two adjacent pressure gauges that captured the first pressure wavefront of an early stimulation stage. During this initial arrival of pressure changes, the direct relationship between the mechanical strain change and pressure response is particularly clear and well-defined. A strong visual coherence between DAS-derived strain rate (blue) and the temporal pressure gradient (red) is evident,

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with positive strain rate (extension) associated with positive pressure gradient (pressure increase). Despite some variability, the general waveform characteristics match well throughout the injection sequence.

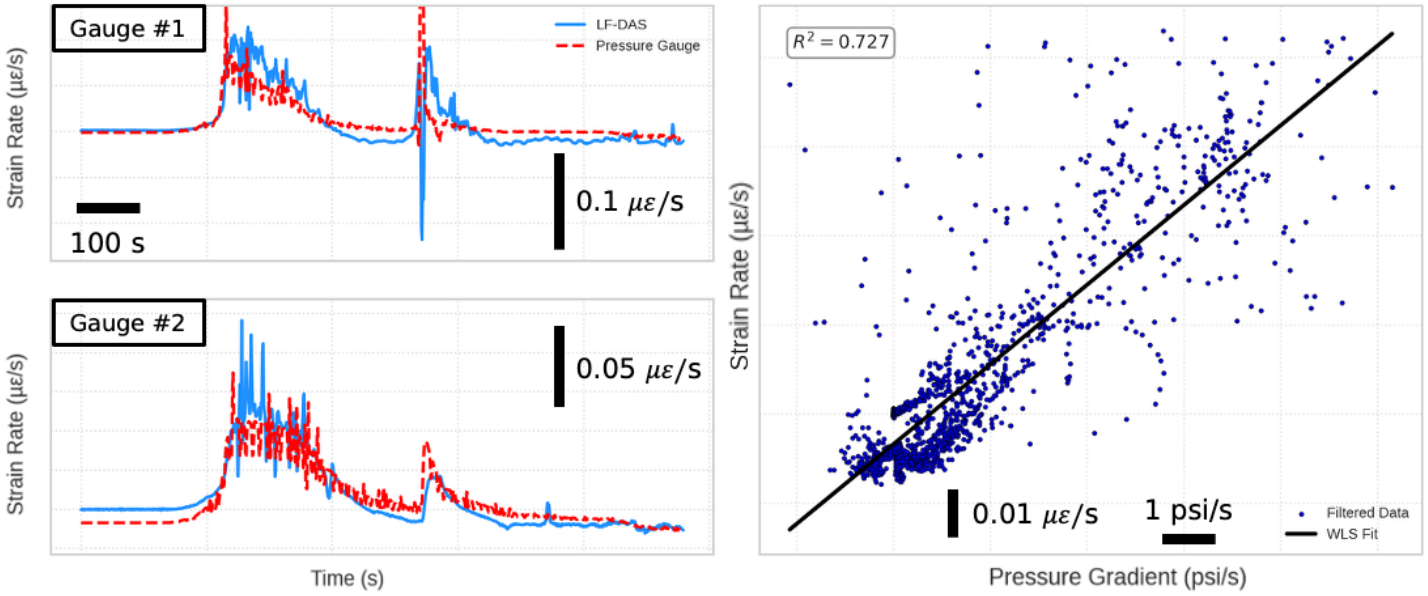


Fig. 3—Correlation between LF-DAS strain rate and data from collocated external pressure gauges. Left panels: Measured LF-DAS strain rate (blue) plotted with strain rate predicted from the temporal gradient of pressure gauge readings (red) at two distinct gauge locations. The prediction uses the linear model derived from the data in the right panel. Right panel: Cross-plot of measured LF-DAS strain rate against the temporal pressure gradient. A Weighted Least Squares (WLS) linear fit (black line) indicates the strong positive linear relationship between these two measurements.

Numerical Modeling and Interpretation

We propose that the pressure communication recorded by the pressure gauges positioned away from the fracture hit sites, as well as the linear migrating signals captured by LF-DAS, result from pressure movement along the monitoring well due to compromised cement quality. The leakage route likely arises from partial debonding at the cement-formation interface or incomplete cement coverage in the annular space. Physically, this debonded zone acts as a pre-existing micro-fracture or a 'planar conduit' that is susceptible to hydraulic activation. Such debonding may be caused or exacerbated by cement shrinkage or air bubbles on the top of horizontal wellbores. This inadequately bonded area enables fluid pressure to propagate along the wellbore annulus, thereby allowing radial stress to affect the cement sheath and the sensing cable.

We interpret the strain response captured by LF-DAS as the result of elastic deformation governed by Poisson's effect. The cement sheath is considered a near-linearly elastic medium, capable of deforming under radial stress/pressure while remaining structurally intact. As shown in Figure 4, fluid pressure enters the debonded annular pathway from the fracture intersection and propagates along the poorly bonded zone between cement and formation. This migrating pressure exerts radial compression on the outer wall of the cement and cable (left panel), generating a mechanical response within the cement body.

The linear fitting results indicate an average ratio of 8.94×10^{-9} $\mu\varepsilon\cdot psi^{-1}$ between the LF-DAS-derived strain rate and the temporal gradient of pressure over multiple injection stages (Figure 3). Assuming a static Young's Modulus E = 30 GPa of the cement sheath, which better represents the bulk material quasi-static deformation measured by LF-DAS, Poisson's ratio can be estimated with:

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$$\nu(x_j, t) = \frac{\dot{\varepsilon}(x_j,t)}{\dot{P}(x_j,t)/E} \quad (2)$$

where $\nu(x_j, t)$ is Poisson's ratio measured at location x_j at time t, ε is the dynamic strain rate from Equation 1, and $P(x_j, t)$ is the temporal pressure gradient from pressure gauge readings at time t. However, the resulting Poisson's ratio is approximately 0.028, which is significantly lower than typical values for most solid materials. A likely explanation is that the cement sheath is still mechanically coupled to the surrounding formation. In this case, when radial pressure increases and the cement attempts to expand laterally, the adjacent rock formation resists that deformation. This lateral confinement reduces the axial strain induced by the Poisson effect, leading to an underestimated strain-to-pressure ratio and, consequently, an unrealistically low inferred Poisson's ratio. Another possibility is that the leakage does not cover the full 360-degree borehole cross-section, which can be caused by air bubbles or cement shrinkage on the high side of the horizontal wellbore. In this case, pressure is not applied uniformly in the radial direction, which would also reduce the efficiency of the Poisson's effect.

Some deviations between the two signals can also be observed (Figures 2, 3). These differences likely arise from additional mechanical responses from fracture propagation captured by DAS—such as near-wellbore deformation and stress shadow—that are not reflected in the pressure gradient measurements. Because LF-DAS captures both pressure-induced strain due to Poisson's effect and fracture-induced strain due to geomechanical deformation, it is expected to exhibit local features that deviate from pressure-only responses.

Through Poisson's effect, the applied radial pressure induces axial extension inside the cement matrix, which includes the embedded fiber-optic cable. This results in a measurable strain along the wellbore, which develops synchronously with the propagating pressure front that is also recorded by external pressure gauges. Together, the radial loading mechanism and the corresponding axial deformation—highlighted across both cross-sectional and longitudinal views in the schematic (Figure 4)—demonstrate a consistent physical process that links pressure migration along a debonded interface to the DAS strain signals observed in the field.

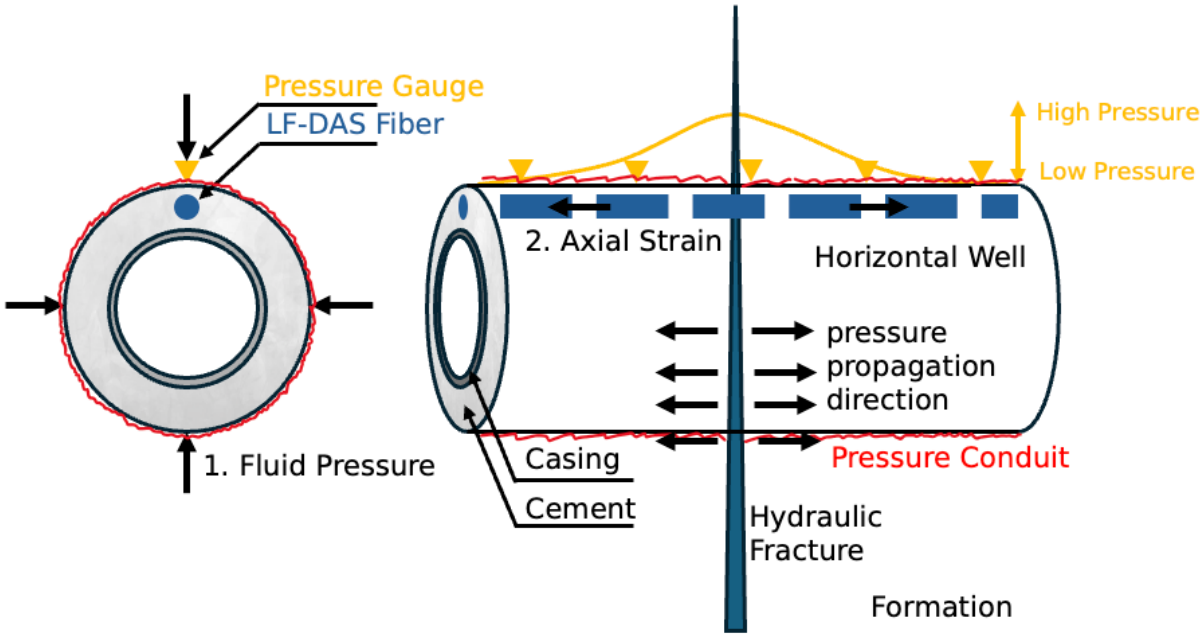


Fig. 4—Schematic illustrating how radial fluid pressure outside the cement sheath leads to axial strain in the cement and fiber. Radial compression from the pressurized fluid induces horizontal stretching through Poisson's effect, causing the observed LF-DAS signals.

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The mechanical coupling described above assumes a relatively simple pressure conduit governed by elastic deformation, which can effectively explain the heel-ward pressure propagation captured by LF-DAS and pressure gauges. However, when the conduit intersects existing conductive hydraulic fractures, the pressure response can deviate from this idealized 1-D behavior. From the borehole or conduit perspective, the connecting hydraulic fractures represent a pressure sink that can absorb a significant amount of fluid without a substantial pressure increase, due to the large storage capacity of the hydraulic fractures. This is equivalent to significantly increasing the effective compressibility at the fracture hit locations, which in turn lowers the apparent hydraulic diffusivity of the conduit.

This adjustment is grounded in the governing equation for hydraulic diffusivity:

$$\eta = \frac{k}{\phi \mu c} \tag{3}$$

where k represents the intrinsic permeability of the conduit, ϕ is the porosity, μ is the dynamic viscosity of the fluid, and c is the effective compressibility. In the context of our model, c is evaluated at the scale of the near-wellbore conduit and defined as:

$$c_{well} = \frac{1}{V_{cell}} \frac{dV}{dp} \tag{4}$$

Here, V_cell represents the small control volume of the monitoring conduit, which is orders of magnitude smaller than the connected fracture network. When the conduit intersects a high-volume fracture, a small pressure perturbation can induce a disproportionately large influx of fluid from the fracture into the conduit. This results in a large dV despite minimal local pressure change (dp), driving the computed c_well to significantly higher values. In turn, this sharply reduces the local diffusivity η, producing a numerically and physically consistent pressure sink at the fracture-hit location.

To verify our conceptual interpretation, a one-dimensional numerical model was developed to simulate fluid pressure migration along the monitoring well. The primary goal of this model is to match recorded LF-DAS and external pressure gauge measurements by simulating the pressure propagation and corresponding strain response. To compare the modeled pressure P(x, t) with the LF-DAS measurements, we leverage the linear relationship established in Figure 3 and Equation 2. Since the strain rate ε is shown to be linearly proportional to the temporal pressure gradient dP/dt via Poisson's effect, the numerical pressure solution can be transformed into a synthetic strain rate field. This allows the 1D diffusion equation—which governs the physical fluid migration—to serve as the forward engine for matching the observed distributed strain signals. In this study, we picked two stimulation stages to demonstrate the data fitting and history matching.

We employed the finite-difference method to simulate single-phase fluid pressure diffusion over the approximately 6,000-ft horizontal wellbore. The mesh was uniformly discretized at a base resolution of 1 ft. To better capture local variations near fracture-hit locations, high-resolution grid refinement was applied locally. Specifically, for each fracture-hit point identified from the LF-DAS field data, the grid spacing was refined five-fold within a ±1 ft window, resulting in 11 grid cells within the 2-ft vicinity surrounding each fracture. At both the heel and toe boundaries, no-flow (Neumann) conditions were imposed—representing, respectively, the well-cemented vertical well section and the tight formation near the toe of the well.

The underlying governing equation is the one-dimensional pressure diffusion equation:

$$\frac{\partial P}{\partial t} = \eta \frac{\partial^2 P}{\partial x^2} \tag{5}$$

where P is the pressure in the pressure pathway, t is time, x is the spatial coordinate along the wellbore, and η is the hydraulic diffusivity.

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This equation assumes slightly compressible single-phase flow in a homogeneous porous medium. The finite-difference method discretizes this PDE in both space and time, enabling efficient numerical propagation of the pressure signal over time. Local source terms can be introduced using pressure or strain measurements at fracture-hit points to simulate fluid injection into the conduit from hydraulic fractures, and spatial variations in η can be used to account for changes caused by fracture connectivity or cement quality degradation.

The initial pressure field was assigned uniformly based on the formation pressure measured by external pressure gauges prior to stimulation. During each stimulation stage, pressure values were prescribed at locations corresponding to fracture hits, with their values being the pressure readings from the nearest external gauges at the monitor well. These real-time field measurements served as proxies for the pressure in the stimulated hydraulic fractures, ensuring that the model reproduced the pressure response behavior observed during the actual operation.

With the numerical framework in place, the next critical step was to determine the appropriate value of hydraulic diffusivity, which is a key parameter that governs the rate and extent of pressure propagation along the well. In this study, due to the dynamic interaction with hydraulic fractures, the diffusivity is treated as a time- and space-dependent variable.

To establish a baseline diffusivity, we selected a representative pressure wavefront from a region unaffected by prior fracture hits (taken from an earlier, lower-noise stage outside the depth range of Figure 6). Using the finite-difference diffusion model mentioned above, we simulated the pressure propagation and compared the results against both external pressure gauge measurements and pressure fields derived from LF-DAS strain data. A diffusivity value on the order of 480 ft^2/s produced a close match to the observed data, accurately reproducing both the timing and amplitude of pressure increases across the sensors. This comparison is shown in Figure 5, where the left panel presents the field data, and the right panel shows the corresponding synthetic pressure field simulated using the baseline diffusivity.

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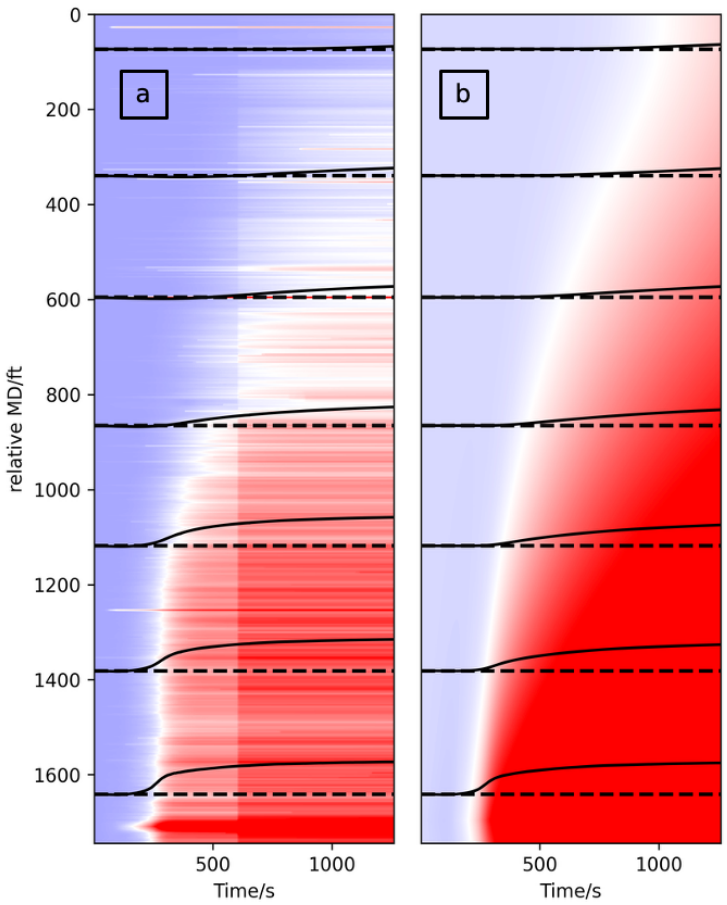


Fig. 5—(a) Measured low-frequency DAS pressure response showing pressure propagation during the beginning of a stimulation stage. Black lines show pressure-gauge readings; dashed lines indicate pressure-gauge locations along the monitor well. (b) Synthetic pressure field computed using a one-dimensional finite-difference diffusion model with an estimated baseline diffusivity of approximately 480 ft²/s.

While the baseline diffusivity provided a good match for heel-ward pressure propagation, it proved insufficient for reproducing pressure behavior in the toe section. During the first stimulation stage shown in the subsequent analysis (Figure 6), although the model captures the overall wavefront propagation, it fails to reproduce the asymmetric pressure migration observed in the field. Specifically, the model assumes uniform diffusivity and does not include any flow barriers that could limit pressure propagation in the toe-ward direction. As a result, it predicts symmetric pressure diffusion around the fracture-hit point, whereas the field data clearly show preferential heel-ward migration and suppressed toe-ward response.

To reproduce this asymmetry, we introduced local reductions in diffusivity at the fracture-hit locations from the previous stage, prior to the start of the subsequent stimulation stage. Maintaining a uniform diffusivity in these zones resulted in overestimated pressure migration and failed to capture the observed shielding behavior, as shown by the first simulated stage in Figure 6. By lowering the local diffusivity at fracture-hit locations in the later stage, the model was able to match both the LF-DAS strain signatures and the external pressure gauge responses in both heel-ward and toe-ward directions for the later stage. The diffusivity reduction was applied only at the first-stage fracture hits. Reproducing the asymmetric response of the second stage requires only the immediately preceding stage to act as a flow barrier. Retaining a uniform baseline diffusivity for the first stage preserves it as a symmetric reference case for direct comparison with the confined response of the second. Because the shielding effect of a debonded conduit is

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established once a single prior stage has been stimulated, a two-stage configuration is sufficient to capture this mechanism.

Interpretation on History Matching Results

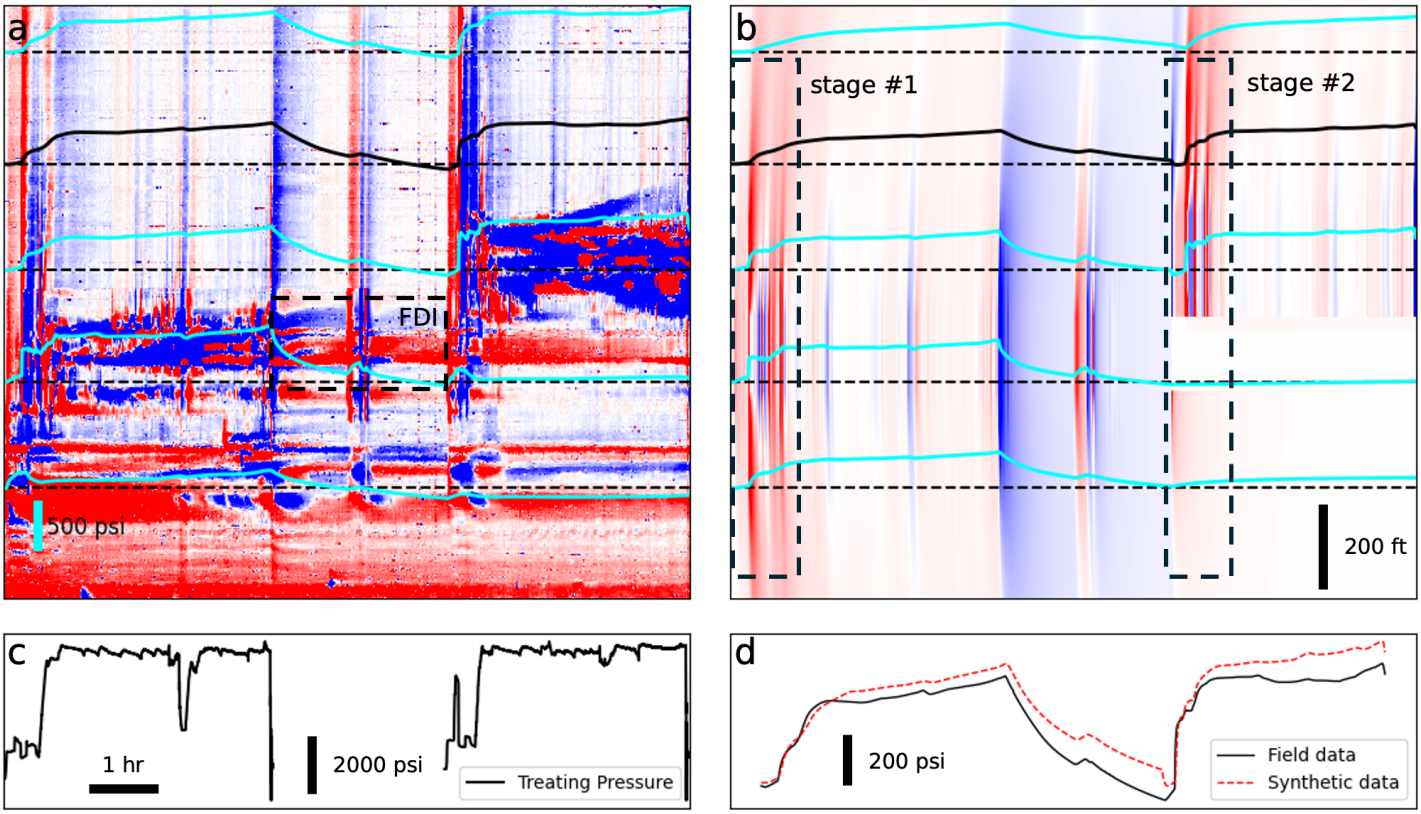


Fig. 6—Comparison between field data and synthetic modeling results for two consecutive stimulation stages. (a) LF-DAS strain-rate waterfall plot with collocated pressure gauge data at the depths indicated by black dashed lines; the dashed box labeled "FDI" marks a zone of strong extensional strain interpreted as fracture-driven interaction (fracture re-opening). (b) Simulated strain-rate and pressure responses, in which Stage 1 is modeled with a uniform baseline diffusivity and serves as a symmetric reference case, while Stage 2 is modeled with locally reduced (adjusted) diffusivity at the previous-stage fracture hits to reproduce the observed asymmetric, heel-ward migration. (c) Treating pressure during the two stages, shown on the same time axis as the LF-DAS waterfall in (a). (d) Comparison between the pressure gauge reading marked in black in (a) and (b).

Figure 6 provides a side-by-side comparison between the field observations and the synthetic model outputs, showing two consecutive stimulation stages. For the first stage modeled, a deliberate discrepancy is highlighted: the model, using a uniform baseline diffusivity, predicts symmetric pressure diffusion around the fracture-hit point. This contrasts sharply with the field data, which clearly shows preferential heel-ward migration and a suppressed toe-ward response. This initial mismatch is intentional, designed to demonstrate that a simple model fails because pre-existing flow barriers from earlier fracture activity were already present in the formation.

In contrast, for the later stage, the simulated pressure and strain responses closely mirror the main features observed in the LF-DAS data and pressure gauges, including pressure front timing, spatial extent, and attenuation across fracture hit zones. This agreement validates the use of spatially varying diffusivity in capturing both the magnitude and directionality of pressure communication during stimulation. To reproduce the observed pressure

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barriers, the diffusivity at previously fractured intervals was reduced by approximately five orders of magnitude—from the baseline value of 480 ft²/s to about 4.8×10^{-3} ft²/s, or 1×10^{-5} of the original value—prior to the onset of the second stimulation stage. This drop ratio was determined through a grid search procedure, in which different levels of diffusivity reduction were systematically tested and compared against the timing and shape of observed pressure and strain wavefronts. The 1×10^{-5} diffusivity reduction produced the best overall match across both LF-DAS and pressure gauge datasets, particularly in reproducing the confined toe-ward propagation pattern. As a pressure diffusion model, it is intended to reproduce only the migrating signal arising from pressure leakage along the annular conduit, not the geomechanical strain associated with fracture opening; mismatches at the fracture-hit locations are therefore expected.

Discussion

Impact of Pressure Drawdown Measurement During Production

The external pressure gauges installed along the monitor well served not only to capture pressure responses during hydraulic stimulation, but also enabled long-term reservoir monitoring throughout the production phase. A key research objective of this field pilot is to understand the geometry of the fracture drainage volume (Raterman et al., 2019). The pressure drawdown recorded at each pressure gauge is assumed to indicate the pressure in the nearby hydraulic fractures (Liang et al., 2022). However, one potential impact of the discovered pressure conduit along the monitor well is to corrupt the pressure drawdown measurement during the production phase.

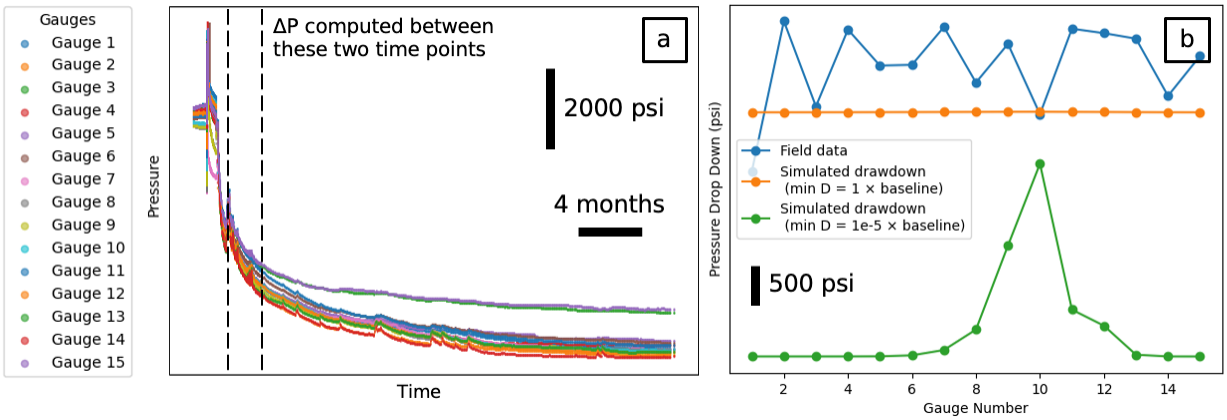


Fig. 7—(a) Pressure profiles recorded by 15 external pressure gauges along the S well over a 30-month production period. The two dashed lines mark the time interval used to compute pressure drawdown (ΔP). (b) Comparison between field-observed pressure drawdown across pressure gauges (blue) and two simulated scenarios: one using the baseline hydraulic diffusivity (orange) shows pressure equalization along the wellbore, while introducing low-diffusivity barriers at fracture intersections results in a highly localized pressure response (green).

As shown in Figure 7a, most of the pressure gauges tracked the decline in production pressure well over time, exhibiting a relatively uniform drawdown profile along the entire length of the monitored wellbore. This uniformity presents a critical interpretational ambiguity, as it could stem from two fundamentally different subsurface scenarios. The first possibility is that the uniform drawdown is an artifact of poor cement quality. In this scenario, a compromised cement sheath creates a high-diffusivity pressure conduit along the well's annulus. This behind-casing flow path would allow pressure changes from only one or a few highly conductive fractures to rapidly equalize along the wellbore. Consequently, all pressure gauges would record a similar, smeared signal, giving the false impression of widespread, uniform drainage. Relying on this interpretation would lead to an overestimation of fracture network connectivity and a skewed understanding of reservoir depletion. Alternatively, the uniform drawdown could be a genuine reflection of a successful stimulation, indicating that many hydraulic fractures are well-connected to the producer and are contributing to production across the well's lateral length. In this case, the cement provides effective zonal isolation,

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and each pressure gauge accurately measures the local reservoir pressure drawdown from nearby, productive fractures. The uniformity would therefore be a true measure of the reservoir's drainage pattern. Without additional diagnostic data, it is impossible to distinguish between these two scenarios. This ambiguity underscores the risk of relying solely on pressure gauge data for long-term production analysis, as it could lead to fundamentally misleading conclusions about both stimulation effectiveness and overall reservoir behavior.

To investigate these possibilities, we performed forward simulations of the production phase. Two scenarios are modeled, as shown in Figure 7b. Our first simulation (orange line) models a scenario with a high-diffusivity annular conduit (poor cement) but assumes all fractures except one are closed and thus present no barriers along the monitor well for pressure communication. This case demonstrates that a single pressure source can indeed produce a uniform drawdown profile that appears similar to the field observation. Our second simulation (green line) models what would happen if open fractures act as low-diffusivity pressure barriers. It shows that a single conductive fracture connecting to the producer in this environment would only create a highly localized pressure drawdown, which is inconsistent with the field data. These two simulations frame the interpretational challenge. The uniform drawdown seen in the field could be explained by two distinct hypotheses: (A) there is only one conductive fracture and the annulus is free of barriers, as modeled by the first simulation; or (B) there are actually many conductive fractures. While each fracture might individually be isolated by surrounding barriers (as in the second simulation), their pressure signals are transmitted and superimposed along the poorly cemented annulus, with the combined effect resulting in the observed uniform drawdown. The ambiguity between these hypotheses is resolved by the DSS measurements from the interference test (Figure 8). The DSS data reveal extensional strain at multiple, discrete locations, confirming the existence of many independently connected, conductive fractures (Dhuldhoya and Friehauf, 2022). This evidence validates hypothesis (B), leading to the conclusion that the observed uniform drawdown is a composite signal created by many active fractures, whose individual pressure responses merge and equalize due to the poor cement quality along the wellbore.

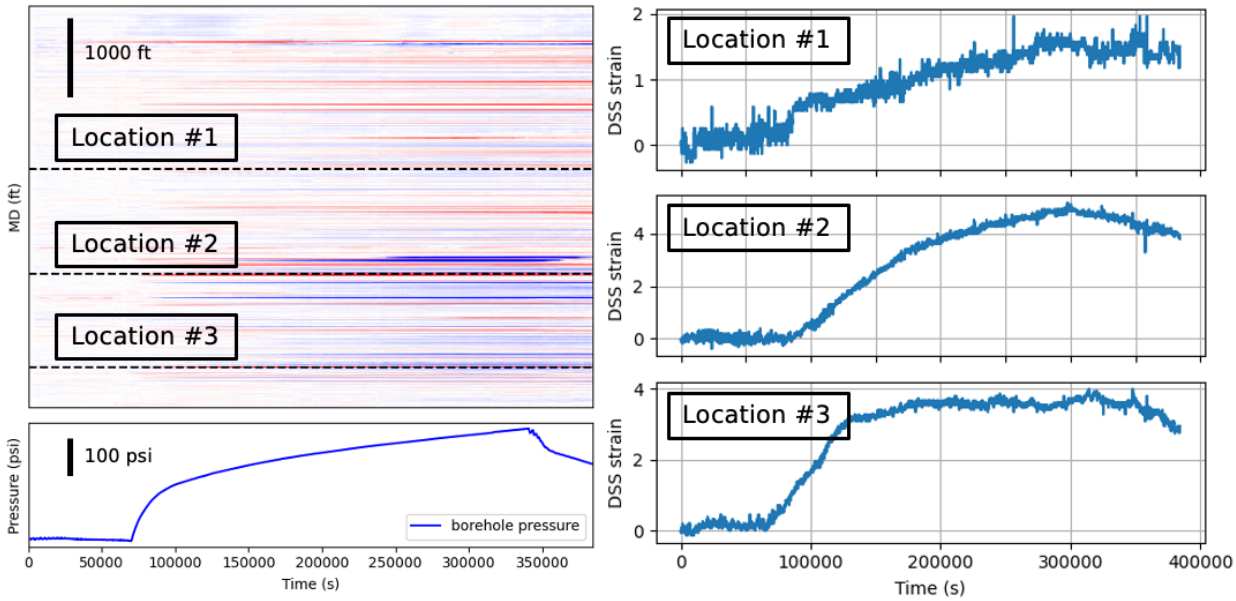


Fig. 8—RFS-DSS response during the interference test. The main waterfall plot (left) shows strain developing over time at multiple locations along the wellbore. Line plots on the right show typical strain responses at the locations marked by dashed lines on the waterfall plot. The pressure profile from the producing well's borehole is shown in the bottom-left panel.

Diagnosing Cement Quality with LF-DAS

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Although the data analysis of this study focuses on dedicated monitoring wells, similar LF-DAS signals have been observed along unconventional producers before their stimulation. This signal has been associated with poor cement quality, with the verification of cement bond logs (Haustveit et al., 2024). Poor cement quality has been shown to compromise stage isolation, leading to overlapping fracture corridors and inefficient stimulation (Mjehovich et al., 2024).

Beyond our field example, such measurements can also be acquired using retrievable wireline (Wu et al., 2020) or disposable fiber (Cox et al., 2025) at significantly lower costs than permanent installations. When detected early through cross-well strain measurements, the diagnosis of poor cement quality provides actionable feedback for optimizing the completion designs of the current monitor wells to mitigate the associated efficiency loss, such as increasing cluster spacing or adjusting stage lengths.

This study establishes the physical connections between the LF-DAS measurement and behind-the-casing pressure communication, which enables quantitative estimation of the diffusivity of the pressure conduit due to poor cement quality. In this context, the external pressure gauges primarily served to ground-truth and confirm that the LF-DAS strain anomalies were indeed caused by annular fluid leakage. Moving forward, external pressure gauges are not required; the fiber-optic cable alone is sufficient to detect and measure the leakage. With proper assumptions about fluid viscosity, porosity, compressibility, and cross-sectional area, it is possible to quantitatively estimate the impact of the pressure conduit on stage isolation with a given cluster spacing, and enable the adjustment of completion designs before the stimulation of the instrumented well.

Furthermore, while the quantitative estimation of hydraulic diffusivity (η) from simulation results allows for an assessment of potential fluid leakage from a current stage to a previous one, the parameters derived from history matching could be used to estimate the initial fluid leakage rate (q) through these compromised zones during stimulation. This leakage rate can be modeled using Darcy's law for flow within the identified pressure conduit along the annulus:

$$q = \frac{k}{\mu}\frac{A}{L}\Delta p = \eta \phi c_t \frac{A}{L}\Delta p \tag{6}$$

where A is the cross-sectional area of the leakage zone, L is the characteristic length of this leakage path, and Δp is the pressure differential driving the flow between stages along the compromised leakage zone. Using the parameters from this study—a hydraulic diffusivity (η) of 480 ft²/s, a calculated cross-sectional area (A) of approximately 0.1 ft², a porosity ϕ of 0.1, a total compressibility c_t of 1×10^{-5} psi^{-1}, and a pressure differential of 2,000 psi—the estimated initial leakage rate (q) for a representative cluster spacing (L) of 16 ft is approximately 0.07 bpm.

Based on this initial estimate, the limited leakage rate might not seem to significantly threaten stage isolation. However, this initial assessment warrants careful consideration of the highly dynamic and aggressive conditions inherent in the hydraulic fracturing process. The high-velocity injection of fluids, especially those laden with abrasive proppant, during the stimulation stages can introduce significant erosional effects on the existing, poorly bonded cement or within incomplete cement coverage areas. This erosion has the potential to progressively enlarge these initial, perhaps minor, leakage pathways, thereby significantly increasing A or k (and thus q) in the Darcy equation. Consequently, what is initially estimated as a manageable, low-rate leakage path for the stimulation period (based on parameters reflecting pre-erosion conditions) could transform into a significant conduit for fluid communication as the treatment progresses. Such an escalation in leakage cannot be ignored, as it could then severely impact stage isolation, lead to inefficient stimulation, and ultimately compromise the objectives of the completion design and overall resource recovery.

Conclusions

In this study, we demonstrated the feasibility of using low-frequency distributed acoustic sensing (LF-DAS) combined with external pressure gauge measurements for quantitatively diagnosing cement quality and associated pressure communication along horizontal wells in unconventional reservoirs. The utilization of LF-DAS strain signals for robust evaluation and characterization of compromised cement sheath integrity offers a significant cost advantage over conventional cement bond logs. Key findings from our analysis include:

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- LF-DAS measurements successfully captured pressure-induced strain responses due to poor cement quality, providing a direct correlation between distributed strain and localized pressure gradients.
- Numerical modeling indicated that fluid pressure migration along compromised cement zones was accurately described by spatially variable hydraulic diffusivity, which was significantly reduced at intersections with previously stimulated hydraulic fractures. This spatial variation in diffusivity allowed precise replication of observed field behaviors.
- Forward modeling highlighted potential pitfalls of relying solely on external pressure gauge data for long-term production interpretation, as poor cement bonding can homogenize pressure responses and lead to misinterpretations of reservoir connectivity. However, ambiguity can be resolved by integrating RFS-DSS measurements, which identify extensional strain at discrete, active fracture locations to confirm true reservoir connectivity.
- Cross-well LF-DAS offers substantial benefits over traditional cement quality diagnostics by providing actionable insights before stimulation. These early insights enable proactive adjustments to completion designs, potentially enhancing overall stimulation efficiency and resource recovery.

The approach presented here lays the foundation for broader applications of LF-DAS in cement integrity assessments. Future work should extend these methodologies to diverse geological settings and further refine quantitative analyses to enhance the predictive capabilities of fiber-optic sensing technologies in wellbore integrity management.

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